Compiler Design Laboratory
Assignment 1 - Regular Expression of a DFA

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Problem Statement

- The regular expression between states i and j through state k

$$
\begin{aligned}
R_{ij}^{k} &= R_{ik}^{k-1}(R_{kk}^{k-1})^{*}R_{kj}^{k-1} \cup R_{ij}^{k-1}, \text{if } k > 0 \\
&= \{a | \delta(q_i, a) = q_j\}, \text{if } k = 0 \wedge i \neq j \\
&= \{a | \delta(q_i, a) = q_j\} \cup \{\epsilon\}, \text{if } k = 0 \wedge i = j
\end{aligned} \tag{1}
$$

- Symbol $a \in \Sigma$; δ is state transition function; i, j, k are non-negative integers
- Consider a Deterministic Finite Automata (DFA) stored in a text file.
- Write a program that will
  - (a) read the file containing DFA from command line argument as input
  - (b) produce the corresponding regular expression as output using equation 1

Input/output format

Input text file format

#states #symbols #init_state_id #transitions #final_states

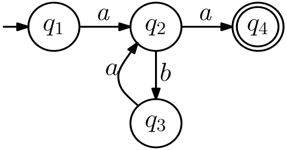
prev_state_id input_symbol_id next_state_id

. . .

prev_state_id input_symbol_id next_state_id

final_state_ids separated by a blank space

For example: Given a DFA



Text file

4 2 1 4 1
1 1 2
2 2 3
3 1 2
2 1 4
4

Output

$a(ba)^*a|aa$

Consider $\Sigma[1] = a, \Sigma[2] = b, \Sigma[3] = c, \ldots$